2. Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to Blur using Gaussian Blur

1. Aim

To write and execute a Python program to read an image and convert it into a blurred image using the Gaussian Blur function in OpenCV.

2. Procedure

1. Import the OpenCV library using import cv2.
2. Read the input image using cv2.imread().
3. Check whether the image is loaded successfully.
4. Apply Gaussian Blur using cv2.GaussianBlur().
5. Display the original image and blurred image using cv2.imshow().
6. Press any key to close the image windows.

3. Code

```
import cv2


# Read the image
img = cv2.imread("img1.jpg")


# Check whether image is loaded
if img is None:
    print("Error: Image not found!")
else:
    # Apply Gaussian Blur
    blur = cv2.GaussianBlur(img, (15, 15), 0)


    # Display original and blurred images
    cv2.imshow("Original Image", img)
    cv2.imshow("Gaussian Blur Image", blur)


    # Save the blurred image
    cv2.imwrite("blurred_image.jpg", blur)
```

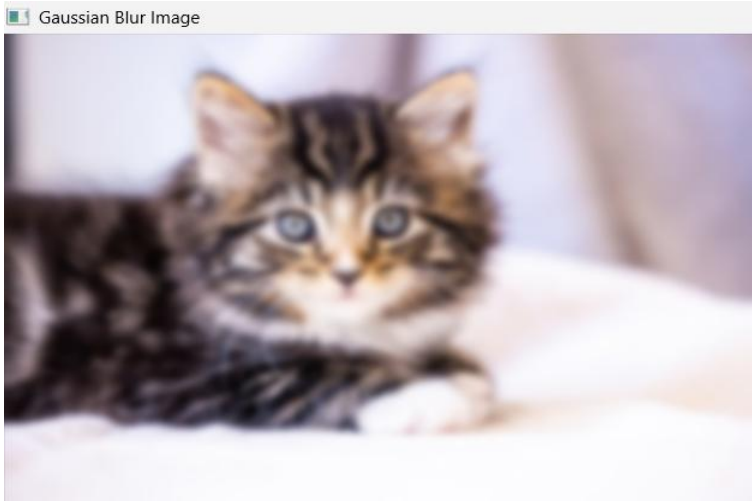
```
cv2.waitKey(0)

cv2.destroyAllWindows()
```

4.Sample input:



5.sample output:



4. Result

Thus, the Python program to read an image and convert it into a blurred image using Gaussian Blur in OpenCV was executed successfully.